MILITARY SPECIFICATION

PRIMER, PERCUSSION, ELECTRIC, M75
METAL PARTS FOR

1. SCOPE

1.1 This specification covers the metal parts for one type primer designated as Primer, Percussion, Electric, M75 for use in Artillery Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-111 - Tape; Paper; Gummed (Sealing and Securing)
QQ-M-151 - Metals; General Specification for Inspection of
PPP-B-636 - Boxes, Fiber
PPP-C-843 - Cushioning Material, Cellulose

MILITARY

MIL-C-490 - Cleaning and Preparation of Ferrous and Zinc Coated Surfaces for Organic Protective Coatings.
MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.
MIL-P-10025 - Packaging and Marking for Domestic Shipment of Inert Ammunition Components.

FSC 1390

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STANDARDS

FEDERAL

Test Method Standard No. 151 - Metals; Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

C-7549256 - Body
A-7549259 - Closure Assembly
B-7549262 - Container Assembly
B-7548483 - Disc, Closing

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
GL-8801950 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawings (dwgs.) C 7549256, A 7549259, B 7549262, B 7548483 and with all requirements specified in applicable specifications.

3.3 Stress relief.-The primer bodies shall be given a stress relief heat treatment of one hour minimum at a temperature of not less than 450° degrees Fahrenheit (°F.) within 24 hours after completion of all manufacturing operations. The test shall be performed as specified in 4.3.1.

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3.4 Mercurous nitrate test.-The primer bodies shall exhibit no cracks when subjected to the mercurous nitrate test specified in 4.3.2.

3.5 Rupture test.-The container cup shall rupture only at the scoring and no pieces shall become detached when the container assembly is subjected to an air or hydrostatic internal pressure of between 1200 and 2400 pounds per square inch (p.s.i.) when tested as specified in 4.3.3.

3.6 Workmanship.-All parts shall be dry and free of chips, dirt, grease, heavy oxide scale and other foreign material. The cleaning method shall not be injurious to any of the parts nor the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

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4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analyses on all material procured directly by the contractor.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.

d. Review of procedures for control and disposition of non-conforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records. (Specification MIL-Q-2550 will apply)

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4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference with operation, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Inspection provisions.

4.2.1 Lot formation.-A lot shall consist of metal parts produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Tube, container (see dwg. B 7549261 covering a detail of dwg. B 7549262).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.25 percent	
101.	Inside diameter at container cup end, maximum (max.)	Gage
Minor:	AQL 0.40 percent	
201.	Depth to chamfer, max	Gage

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4.2.2.2 Container, assembly (see dwg. B 7549262).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 0.65 percent	
101.	Concentricity of outside diameter with threads	Gage
102.	Pitch diameter of thread minimum (min.)	Gage
103.	Major diameter of thread min	Gage
104.	Outside diameter	Gage
105.	Crack or deformation	Visual
106.	Braze incomplete	Visual
Minor:	AQL 4.00 percent	
201.	Total length	Gage
202.	Pitch diameter of threads, max	Gage
203.	Major diameter of threads, max	Gage
204.	Width of slots, min	Gage
205.	Depth of slots	Gage
206.	Depth to container cup, max	Gage
207.	Finish improper	Visual
208.	Evidence of poor workmanship (see 3.6)	Visual

4.2.2.3 Body (see dwg. C 7549256).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	AQL 1.00 percent	
101.	Concentricity of thread with outside surface	Gage
102.	Concentricity of hole with outside surface	Gage
103.	Concentricity of igniter cavity diameter with outside surface	Gage
104.	Concentricity of flange with outside surface	Gage
105.	Concentricity of thread pilot diameter with outside surface	Gage
106.	Pitch diameter of threads	Gage
107.	Minor diameter of thread, max	Gage
108.	Length of datum diameter	Gage
109.	Thickness from base to bottom of igniter cavity	Gage
110.	Thickness of flange	Gage

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Minor: AQL 1.50 percent

201.	Total length	Gage
202.	Length to thread, max	Gage
203.	Length to end of thread, min	Gage
204.	Evidence of poor workmanship (see 3.6)	Visual

4.2.3 Testing.

4.2.3.1 Stress relief (see 3.3) Major defect.-The Government inspector shall monitor this operation as specified in 4.3.1.

4.2.3.2 Mercurous nitrate test (see 3.4) Major defect.-The sampling plans for this test shall be in accordance with MIL-STD-105, using Table IV-A, code letter G and an AQL of 0.65 percent. The test shall be performed as specified in 4.3.2.

4.2.3.3 Rupture test (see 3.5) Major defect.-The sampling plan for this test shall be in accordance with MIL-STD-105, using Table IV-A, code letter K and an AQL of 0.10 percent. The test shall be performed as specified in 4.3.3.

4.2.3.4 Air pressure test (see dwg. 7549262).-The sampling plan for this test shall be in accordance with MIL-STD-105, using code letter L and an AQL of 0.25 percent. The test shall be performed as specified in paragraph 4.3.4.

4.2.3.5 Freedom from alkalies and strong acids.-Three sampling containers from each four hour production shall be selected for this test. The test shall be performed as specified in paragraph 4.3.5.

4.2.3.6 Freedom from oil and grease.-Three sample containers from each four hour production shall be selected for this test. The test shall be performed as specified in 4.3.6.

4.2.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.2 and 4.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring and testing equipment identified by Index of Inspection Equipment Lists No. GL-8801950, or approved equivalents thereto. When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.3 Test methods and procedures.

4.3.1 Stress relief.-The monitoring shall consist of surveillance of the operations at regular but unscheduled intervals to determine that they are being conducted in accordance with the prescribed procedures. When, the inspector discovers that the operations are not being performed in accordance with the required procedures, he shall reject all items stress relieved since his last satisfactory surveillance. The contractor shall reprocess these rejected items 100 percent.

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4.3.2 Mercurous nitrate test.-The primer bodies shall be tested in accordance with Federal Test Method Standard No. 151, Method 831. Each body shall be examined for cracks.

4.3.3 Rupture test.-The container assembly shall be tested in accordance with the procedure specified in the Index of Inspection Equipment List (see 6.2).

4.3.4 Air pressure test.-The sample container shall be subjected to the test and procedure specified in the Index of Inspection Equipment Lists (see 6.2).

4.3.5 Freedom from alkalies and strong acids.-The container assemblies shall be tested in accordance with Specification MIL-C-490. If the limits specified herein have not been complied with, inspection for acceptance shall be stopped until corrective action has been taken. After corrective action, sampling shall be resumed as specified in 4.2.3.5.

4.3.6 Freedom from oil and grease.-The container assemblies shall be tested and action taken in accordance with Specification MIL-C-490.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.

5.1.1.1 Disc, closing (dwg. B-7548483).-Ten thousand closing discs shall be packed in bulk in a 5 by 7 1/2 inch commercial, minimum 30 pound (lb.) basis weight, kraft coin envelope with gummed end flaps.

5.2 Packing.-All parts shall be packed in accordance with the requirements specified in Specification MIL-P-10025 and shall comply with the Quality Assurance provisions thereof.

5.2.1 Level C.-The inert parts shall be packed as indicated below. The construction of the carton and the material requirements of the carton and partitions shall comply with Federal Specification PPP-B-636. The parts manufacturer shall be furnished preproduction samples of all packing material so that proper fit of packing materials and metal parts can be determined. Minor modifications of packing material dimensions to provide a proper fit of all parts, as determined to be necessary by examination of sample packs, is permissable.

5.2.1.1 Body (dwg. C-7549256).-One thousand bodies shall be packed in a 25 by 20 arrangement within half-slotted "A" flute, 200 lb. test, double-faced corrugated partitions in 2 layers in a 275 lb. test RSC style, "B" flute double-faced Type I, class 1, fiberboard carton. The inside dimensions of the carton shall be 15 9/16 by 12 7/16 by 4 3/8 inches. The outside dimensions of the partitions shall be 15 1/2 by 12 3/8 by 1 15/16 inches to form 500 equal cells. A plain pad constructed of the same material as the cartons shall be placed on the bottom, top and between layers. The pad shall be equal to the length and width of the carton less 1/16 inch.

5.2.1.2 Container assembly (dwg. B-7549262).-Five thousand container assemblies shall be packed in a 25 by 40 - 5 layer arrangement within individual half-slotted partitioned cells positioned within a style RSC, 275 lb. test, carton. The carton and partitions shall be constructed of 275 lb. test, Type I, Class 1 fiberboard. The inside dimensions of the carton shall be 21 11/16 by 13 1/2 by 5 15/16 inches.

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The outside dimensions of the partitions shall be 21 5/8 by 13 7/16 by 1 inch to form 1000 equal cells. A "B" flute double-faced corrugated fiberboard pad 21 5/8 by 13 7/16 inches shall be placed beneath the partitions and on top as required to insure a tight pack.

5.2.1.3 Disc, closing (dwg. B-7548483).-One or more envelopes containing closing discs shall be packed in a RSC style "B" flute, double-faced, corrugated fiberboard carton. Inside dimensions of the carton shall be 10 by 10 by 5 inches. Dunnage conforming to Specification PPP-C-843 shall be used to fill voids.

5.3 Sealing.-The cartons shall be closed in accordance with the requirements specified in Specification MIL-P-10025.

5.4 Percentage of extra parts packing.-When feasible, for small and light weight parts, the entire packing lot quantity, including the percentage of extra parts, shall be packed in the prime containers.

5.5 Shipping.-When component parts from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

5.6 Marking.-Marking of the cartons and envelopes shall comply with Specification MIL-P-10025.

5.7 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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